Cooldown study: which 40-step lookahead ranks interval candidates like the final result?

Built 2026-10-16 13:03 PT. Experiment design by Roland Gao; runs and analysis by Claude. Code: research/cooldown_search_probe.py (runs), research/cooldown_study.py (analysis), research/make_cooldown_study.py (this PDF). Logs: logs/research_cooldown_search_probe_*.

Question

An interval search picks each interval's LR by training the interval with a candidate LR, then a short lookahead (the cooldown), and scoring the result. Which cooldown ranks the candidates the way the final accuracy does? Compared, all with a 40-step lookahead (one interval at batch size 2000): no lookahead; linear decay to 0 (conv only, or all four groups); one searched constant cooldown, either for conv only with the other groups on their ground-truth schedules, or shared by all four groups; and four searched constant cooldowns (one per group).

Setup

Item	Value
Model / data	CIFAR-10 airbench, batch size 2000, 200 steps, baseline_bs2000 (conv Muon2 0.22 -> 0; head, norm_bias, whiten_bias SGD on their baseline schedules)
Intervals	4 intervals of 40 steps: steps 0-40, 40-80, 80-120, 120-160 (the last interval has no room for a 40-step lookahead)
Candidates	7 constant conv LRs for the interval: the 0.6^k grid point nearest the baseline's mean conv LR in that interval, and 3 grid steps either side; other groups on the baseline
Ground truth	Final test TTA when training continues on the baseline schedule after the interval
No lookahead	Test TTA right after the interval
Linear to 0	Conv only: conv decays linearly from the candidate to 0 over 40 steps, the other groups continue their schedules. All groups: every group decays linearly to 0 from its current value
1 searched cooldown: conv only	Conv constant at m x candidate for 40 steps; the other groups follow their baseline (ground-truth) schedules; m chosen from {1, 0.6, 0.36, 0.22, 0.13, 0.078, 0.047, 0} by the best score on that seed
1 searched cooldown: shared	Constant LRs for 40 steps: conv at m x candidate, each other group at m x its mean LR over the interval; one m for all groups, chosen as above
4 searched cooldowns	As above but one m per group; coordinate search from the best shared m, one group at a time over the same 8 values, until no group changes (at most 3 passes)
Seed	1 training seed (0); every method uses the same prefix and data order
Score	Test-set TTA accuracy (%)

Results

Picked LR = the candidate with the highest score. Regret = ground truth at its best candidate minus ground truth at the picked one (pp). One seed: single-run noise is about 0.13 pp, so differences of that size between candidates are not resolved.

Picked LR (regret, pp)

Method	Interval 1	Interval 2	Interval 3	Interval 4
Ground truth (final TTA)	0.078 (0.00)	0.13 (0.00)	0.078 (0.00)	0.078 (0.00)
No lookahead	0.078 (0.00)	0.028 (0.34)	0.028 (0.01)	0.028 (0.29)
Linear to 0: conv only	0.13 (0.05)	0.078 (0.26)	0.078 (0.00)	0.028 (0.29)
Linear to 0: all four groups	0.13 (0.05)	0.078 (0.26)	0.047 (0.21)	0.028 (0.29)
1 searched cooldown: conv only, others on ground truth	0.13 (0.05)	0.078 (0.26)	0.047 (0.21)	0.047 (0.21)
1 searched cooldown: shared by all four groups	0.13 (0.05)	0.078 (0.26)	0.047 (0.21)	0.047 (0.21)
4 searched cooldowns (one per group)	0.13 (0.05)	0.078 (0.26)	0.047 (0.21)	0.047 (0.21)

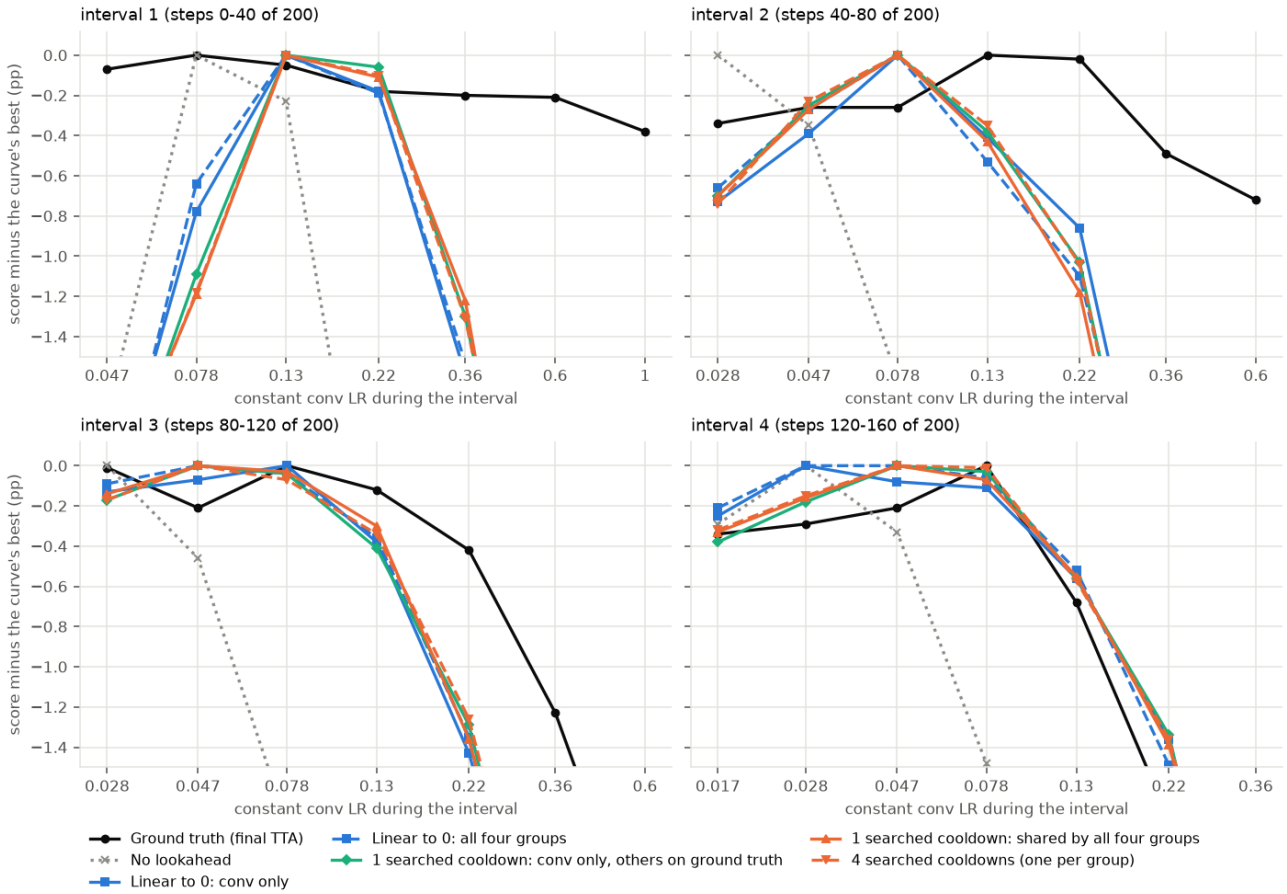


Figure 1. Each method's score per candidate, relative to its own best (the point at 0 is its pick).

Interval 1: score per candidate conv LR (bold: each row's best)

Method	0.047	0.078	0.13	0.22	0.36	0.6	1
Ground truth (final TTA)	93.99	**94.06**	94.01	93.88	93.86	93.85	93.68
No lookahead	81.57	**83.35**	83.12	80.45	68.80	56.07	35.18
Linear to 0: conv only	89.04	90.59	**91.37**	91.19	89.78	87.38	84.39
Linear to 0: all four groups	88.85	90.63	**91.27**	91.08	89.74	87.41	84.10
1 searched cooldown: conv only, others on ground truth	88.58	89.83	**90.92**	90.86	89.62	87.45	85.95
1 searched cooldown: shared by all four groups	88.65	89.70	**90.89**	90.78	89.67	87.16	85.13
4 searched cooldowns (one per group)	88.79	89.89	**91.07**	90.97	89.76	87.65	86.13

Candidate	Conv only: chosen m (seeds)	Shared: chosen m (seeds)	4 searched: chosen (conv, head, norm_bias, whiten_bias) (seeds)
0.047	0.6 x1	0.6 x1	(1, 1, 0.6, 0.36) x1
0.078	0.36 x1	0.36 x1	(0.36, 1, 1, 0.22) x1
0.13	0.22 x1	0.22 x1	(0.22, 1, 0.22, 0.22) x1
0.22	0.22 x1	0.22 x1	(0.22, 0.36, 0.22, 1) x1
0.36	0.13 x1	0.13 x1	(0.13, 0.047, 0.13, 0.13) x1
0.6	0.078 x1	0.078 x1	(0.078, 1, 0.36, 0.078) x1
1	0.078 x1	0.078 x1	(0.078, 0.6, 0.078, 0.6) x1

Interval 2: score per candidate conv LR (bold: each row's best)

Method	0.028	0.047	0.078	0.13	0.22	0.36	0.6
Ground truth (final TTA)	93.76	93.84	93.84	**94.10**	94.08	93.61	93.38
No lookahead	**91.12**	90.77	89.51	87.43	82.45	71.27	57.08
Linear to 0: conv only	92.25	92.59	**92.98**	92.57	92.12	90.18	87.94
Linear to 0: all four groups	92.27	92.66	**92.93**	92.40	91.83	90.11	87.78
1 searched cooldown: conv only, others on ground truth	92.16	92.61	**92.86**	92.48	91.83	89.82	87.93
1 searched cooldown: shared by all four groups	92.18	92.61	**92.88**	92.45	91.70	89.73	87.84
4 searched cooldowns (one per group)	92.20	92.71	**92.94**	92.59	91.90	89.87	87.98

Candidate	Conv only: chosen m (seeds)	Shared: chosen m (seeds)	4 searched: chosen (conv, head, norm_bias, whiten_bias) (seeds)
0.028	0.6 x1	0.6 x1	(0.6, 0.6, 1, 0.6) x1
0.047	0.36 x1	0.36 x1	(0.36, 1, 0.047, 0.36) x1
0.078	0.22 x1	0.22 x1	(0.22, 0.13, 0.22, 0.22) x1
0.13	0.22 x1	0.22 x1	(0.22, 1, 0.36, 0) x1
0.22	0.13 x1	0.13 x1	(0.13, 1, 1, 0.078) x1
0.36	0.13 x1	0.13 x1	(0.13, 0.6, 0, 0.047) x1
0.6	0.078 x1	0.078 x1	(0.078, 0.047, 0.13, 0.078) x1

Interval 3: score per candidate conv LR (bold: each row's best)

Method	0.028	0.047	0.078	0.13	0.22	0.36	0.6
Ground truth (final TTA)	93.99	93.79	**94.00**	93.88	93.58	92.77	91.54
No lookahead	**92.59**	92.13	90.81	87.68	82.32	71.78	56.20
Linear to 0: conv only	93.26	93.32	**93.39**	93.01	91.96	90.45	87.63
Linear to 0: all four groups	93.28	**93.37**	93.33	93.01	92.02	90.23	87.59
1 searched cooldown: conv only, others on ground truth	93.16	**93.33**	93.29	92.92	92.04	90.31	88.61
1 searched cooldown: shared by all four groups	93.13	**93.27**	93.24	92.97	91.91	90.29	88.64
4 searched cooldowns (one per group)	93.18	**93.35**	93.28	93.01	92.09	90.44	88.76

Candidate	Conv only: chosen m (seeds)	Shared: chosen m (seeds)	4 searched: chosen (conv, head, norm_bias, whiten_bias) (seeds)
0.028	0.36 x1	0.22 x1	(0.22, 0.13, 1, 0.22) x1
0.047	0.22 x1	0.36 x1	(0.22, 1, 1, 0.36) x1
0.078	0.22 x1	0.22 x1	(0.22, 1, 0.36, 0.22) x1
0.13	0.22 x1	0.22 x1	(0.22, 1, 0.22, 0.22) x1
0.22	0.22 x1	0.13 x1	(0.22, 0.36, 1, 0.13) x1
0.36	0.13 x1	0.13 x1	(0.13, 0.078, 0.6, 0.13) x1
0.6	0.078 x1	0.13 x1	(0.078, 0.36, 0.13, 0.13) x1

Interval 4: score per candidate conv LR (bold: each row's best)

Method	0.017	0.028	0.047	0.078	0.13	0.22	0.36
Ground truth (final TTA)	93.65	93.70	93.78	**93.99**	93.31	92.23	89.58
No lookahead	92.89	**93.18**	92.85	91.70	88.84	83.30	72.23
Linear to 0: conv only	93.55	**93.80**	93.72	93.69	93.24	92.44	90.63
Linear to 0: all four groups	93.55	**93.76**	**93.76**	93.70	93.24	92.27	90.57
1 searched cooldown: conv only, others on ground truth	93.51	93.71	**93.89**	93.86	93.33	92.55	90.70
1 searched cooldown: shared by all four groups	93.56	93.73	**93.89**	93.82	93.34	92.50	90.89
4 searched cooldowns (one per group)	93.62	93.79	**93.94**	93.93	93.37	92.58	90.93

Candidate	Conv only: chosen m (seeds)	Shared: chosen m (seeds)	4 searched: chosen (conv, head, norm_bias, whiten_bias) (seeds)
0.017	0.6 x1	0.6 x1	(0.6, 1, 1, 0.6) x1
0.028	0.6 x1	0.6 x1	(0.6, 1, 0.36, 0.6) x1
0.047	0.36 x1	0.36 x1	(0.36, 0.22, 1, 0.36) x1
0.078	0.22 x1	0.22 x1	(0.22, 0.047, 0.22, 0.22) x1
0.13	0.13 x1	0.13 x1	(0.13, 0.22, 0.13, 0.13) x1
0.22	0.13 x1	0.13 x1	(0.13, 1, 0.078, 0.13) x1
0.36	0.13 x1	0.13 x1	(0.13, 0.078, 1, 0.13) x1